



Topical hemostatic agents for malignancy-related GI bleeding: Is it time to prioritize their use?

Success is not about perfection. It is about progress!

Gastrointestinal bleeding can be the initial manifestation or a complicating problem during the follow-up of patients with GI cancer. Over the past 2 decades, the survival of patients with many solid tumors, including those in the GI tract (esophagus, stomach, colon, rectum, and pancreas) has been increasing at all stages of malignancy.¹ Advances in cancer treatment modalities have been associated with this survival benefit. As a result, the incidence of GI bleeding from GI tumors, both primary and metastatic, is expected to increase. Limited data are available about the frequency, natural history, and treatment outcomes for patients with bleeding associated with GI cancer. In a recent retrospective analysis of the National Inpatient Sample Data Base,² 9.5% of patients with GI cancer had GI hemorrhage. In another retrospective review, 5% of patients with GI tumors underwent endoscopy because of GI bleeding.³ In many patients, the aim of intervention to stop bleeding is to provide a bridge until other more definitive therapies are effective or possible.

As most gastroenterologists know from experience, bleeding from GI tumors is challenging to manage endoscopically. This is because of the frequent absence of a discrete bleeding vessel, the presence of diffuse oozing from the abnormal tumor tissue, and the high rebleeding rate after successful initial hemostasis with the use of standard endoscopic methods.

Topical hemostatic agents (THAs) are an appealing treatment in this setting. They are easy to administer quickly over a large surface area, with a resulting hemostasis even when the bleeding vessel/vessels is/are not visible. Several products are available in the United States and around the world. The use of THAs in ulcer/nontumor bleeding hemostasis has been associated with high initial hemostasis rates. However, rebleeding rates as high as 30% to 50% have also been reported.^{4,5} More recent data however, suggest that rebleeding rates are lower and are similar to those with standard endoscopic therapy, with even lower rebleeding rates in Forest Ib lesions.⁶

In this issue of *Gastrointestinal Endoscopy*, Karna et al⁷ provide much needed data in this setting. The authors conducted a systemic review and meta-analysis of the use of

THA in bleeding from GI tumors. The majority of patients received TC-325 (Hemospray, Cook Medical, Winston-Salem, NC). The primary outcome was primary hemostasis defined by 6 studies as no rebleeding after 1 to 5 minutes of observation after therapy. Secondary outcomes were rebleeding (early, delayed, and aggregate) and all-cause and GI bleeding-related mortality. Early rebleeding was defined in 3 studies as that occurring within 72 hours. The analysis included 16 studies (2 randomized controlled trials and 14 observational) with a total of 530 patients. Two studies were published in abstract form. The majority

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of patients included had oozing bleeding (85.95%) from gastric tumors (60.15%). THA was used as monotherapy in 76.2%. The rate of primary hemostasis was 94.1%. The incidence of early, delayed, and aggregate rebleeding was 13.9%, 11.4%, and 24.2%, respectively. The incidence of GI bleeding-related mortality was 5.9% and of all-cause mortality was 33.1%.

Two important conclusions are evident from this study: high initial hemostasis and a reduction in rebleeding rate in comparison with standard endoscopic interventions.

Keeping in mind the methodologic limitations of the individual studies included and the inherent limitations of meta-analyses themselves, the data provided help further our understanding of the role of THA in achieving hemostasis among patients with bleeding from GI tumors. THA use was associated with primary hemostasis in 95% of patients, which is similar to that with ulcer bleeding.³⁻⁵ Overall, the rebleeding rate was 24%, half of which occurred in the first 3 days after treatment. This rebleeding rate is much lower than rebleeding rates associated with the use of standard therapy in this setting and provides a much-needed improvement and potential bridge until

more definitive therapy is used. This lasting benefit occurred even though the gel applied falls out within 24 hours in most patients and suggests that THA may provide a favorable environment for a stable clot/hemostasis, allowing for longer duration of benefit. This reprieve can also be used to initiate or re-evaluate the goals of care discussions with the patient. It is also encouraging that despite a rebleeding rate of 24%, the bleeding-related mortality rate was 5.9%. In other words, 75% of rebleeding episodes were not fatal and thus presumably were successfully controlled. It would be interesting to know how hemostasis was achieved in those rebleeding episodes. Specifically, it would be interesting to know whether the repeated use of THA achieves the same hemostasis success rate as during the index episode and whether there is any subpopulation of patients with tumor bleeding who would benefit from retreatment with THA. The overall mortality rate of 33% is lower than reported in this group of patients. However, it is difficult to know whether it is a benefit of treatment or whether it is reflective of either a short follow-up period or the selection of patients with longer life expectancy. Other studies have also reported better survival in similar patients undergoing endoscopy.³

Overall, the findings of this study should be interpreted in the specific context of the study limitations. First and foremost, only 2 studies included were randomized controlled studies. Even then, both were small (10 and 18 patients in the intervention arm, respectively), were pilot rather than complete studies, and included patients from 2 centers and a single center, respectively. One of these studies was published in an abstract form. Therefore, the risk of bias from confounding by indication remains high. This limitation cannot be offset by the higher number of patients that aggregating 16 studies achieves. Second, as the authors report, the funnel plot was asymmetric, which indicates the presence of publication bias. Publication bias in turn indicates that the rates presented in the study are an overestimation of the true rates. Combined with the first limitation, this limitation highlights the need for large, multicentered, controlled, and adequately powered studies before THA can be strongly recommended as a preferred treatment modality in clinical practice. Third, THA was used as a second-line treatment in 25% of the studies included. Whereas the intent of including these patients is understandable (increasing the number of patients), the heterogeneity of the resulting study population makes the interpretation of the results challenging. Fourth, information about the use of chemotherapy, radiation therapy, surgery or a combination during the follow-up period is not available. Ideally, patients who receive such treatment, especially surgical tumor resection, should be excluded or at least censored from the study on the date of this event in an effort to avoid overestimating the long-term hemostasis rate. Fifth,

inherent in the meta-analysis methodology, the length of the follow-up period and the type of outcome ascertainment (active vs passive) was not standardized or well known in the studies included. This is especially relevant for delayed rebleeding outcome because patients with GI tumors might elect for hospice rather than seek medical care after a rebleeding episode. Finally, it is important to situate the results obtained in the context of cost of care. In the current healthcare environment, where efficiency of healthcare delivery, comparative effectiveness, and cost effectiveness are ever more important, information on the cost of THA treatment and its impact on additional treatment and cost would be essential for resource allocation and for decision making at the patient, healthcare, and societal levels.

Bleeding from GI tumors is becoming increasingly prevalent, and finding effective treatment modalities that can achieve hemostasis in this setting is key in view of the limited success rate of the traditional endoscopic treatment modalities. The current study by Karna et al⁷ is a much-needed summary of the existing data. It highlights the potential of such a treatment modality, but it also highlights the lack of high-quality data upon which to justify the use of THA in clinical practice. In the meantime, endoscopists involved in the care of such patients can use the current study data to supplement their therapeutic decisions and consider using THA as their primary therapy. Large randomized controlled studies are needed to fully clarify the utility of this promising technology in GI tumor bleeding.

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Abbreviation: THA, topical hemostatic agent.

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